

## AI Assistant Architecture (Full Starter Template)

Website Type: Full AI Assistant (chat, images, files, voice)

AI Model: Recommended option

Technology: HTML/CSS/JavaScript + Python

Hosting: Free hosting

### Recommended Architecture

#### Frontend

- HTML
- CSS
- JavaScript

#### Backend

- Python + Flask

#### AI Services

- Google Gemini API
- Hugging Face APIs
- Browser Speech Recognition + Speech Synthesis

#### Database

- SQLite

#### Features

- AI Chat
- Voice Input/Output
- Image Generation
- File Upload (PDF, TXT)
- Chat History
- Mobile-Friendly UI
- Dark Mode

### Project Structure

```
ai-assistant/  
├──  
│   ├── app.py  
│   ├── requirements.txt  
│   └── database.db  
├──  
│   ├── static/  
│   │   ├── style.css  
│   │   └── script.js  
│   └── templates/  
│       └── index.html  
└── uploads/
```

#### Main Backend (app.py)

```
from flask import Flask, render_template, request, jsonify  
import google.generativeai as genai  
...
```

#### Frontend (index.html)

```
<!DOCTYPE html>  
<html>  
<head>  
<title>AI Assistant</title>  
</head>  
<body>  
...  
</body>  
</html>
```

#### JavaScript

```
async function sendMessage() {  
...  
}
```

#### Deployment:

- Render
- Railway
- GitHub